Practice Problems for Chapters 5.1-6.5

Chapter 5

1. Find all the intervals where $f(x)$ is increasing, decreasing, concave up, concave down.
   (1) $f(x) = \frac{1}{4}x^4 - 2x^3 + \frac{9}{2}x^2 + 5$
   (2) $f(x) = \dfrac{x}{x^2+3}$
   (3) $f(x) = \ln \sqrt[3]{x^2+4}$
   (4) $f(x) = xe^{x^2}$
   (5) $f(x) = \sin x - \cos x, \quad x \in [-\pi, \pi]$

2. Given a function $f(x)$, (a) Find all critical points of f, (b) Find all relative extrema of f using **the first derivative test** or **the second derivatives test**.
   (1) $f(x) = 2x^3 + 3x^2 - 12x + 5$
   (2) $f(x) = 2x + 2x^{2/3}$
   (3) $f(x) = (x-3)e^x$
   (4) $f(x) = \dfrac{x+3}{x-2}$
   (5) $f(x) = |3x - x^2|$

3. Find the absolute maximum and minimum value of f on the given interval if any.
   (1) $f(x) = 2x^3 - 3x^2 - 12x, \quad [-2, 1]$
   (2) $f(x) = \sin x - \cos x, \quad [-\pi/4, \pi/2]$
   (3) $f(x) = \dfrac{3x}{\sqrt{4x^2+1}}, \quad [-1, 1]$
   (4) $f(x) = 2x^3 - 9x + 1, \quad (-\infty, +\infty)$
   (5) $f(x) = \dfrac{x^2+1}{x+1}, \quad (-1, 5)$

4. A rectangle field is to be bounded by a fence on four sides. Find the dimensions of the field with maximum area that can be enclosed with 1000 feet of fence. (Hint: Formulate the problem as a max/min problem on an approximate interval)

5. Find a point on the curve $x^2 + y^2 = 1$ that is closest to the point $(2, 1)$.

6. Approximate the real solution of the following equations using the Newton's method as the examples in Section 5.6.
   (1) $x^5 + x^4 - 5 = 0$
   (2) $\sin x = x^2$
   (3) $e^{-x} = \ln x$

7. State the Rolle's Theorem and the Mean-Value Theorem.

8. Let $s(t) = \dfrac{100}{t^2+12}$ be the position of a partical moving along a coordinateline, where s is in feet and t is in seconds ($t \geq 0$).
   (a) Find the velocity of this particle $v(t)$.
   (b) Find the acceleration of this particle $a(t)$.
   (c) At what time does the partical attain it maximum speed.

Chapter 6.1-6.5

1. Evaluate the following integrals.
   (1) $\int (x + x^5 - x^{-2} - 4x^{1/3})\ dx$
   (2) $\int \sec x \tan x\ dx$
   (3) $\int (\frac{1}{2t} + 3\sqrt{t} + 2\cos t - \sqrt{2}e^t)\ dt$
   (4) $\int \frac{1}{x^2+1}\ dx$
   (5) $\int \frac{1}{\sqrt{1-x^2}}\ dx$

2. Evaluate the following integrals using appropriate substitutions.
   (1) $\int \sin(7x+1)\ dx$
   (2) $\int 2x(x^2-2)^{23}\ dx$
   (3) $\int \frac{2}{x \ln x}\ dx$
   (4) $\int (1+\sin t)^9 \cos t\ dt$
   (5) $\int \frac{e^x}{1+e^x}\ dx$
   (6) $\int x^3\sqrt{5+x^4}\ dx$
   (7) $\int \frac{y}{\sqrt{2y+1}}\ dy$
   (8) $\int \frac{1}{\sqrt{1-4x^2}}\ dx$

3. Use the Definition 6.4.3 with x_k^* as the left endpoint of each equal subinterval to find the area under the curve $y = 9 - x^2$ over the interval $[0, 3]$.

4. Express the limit
$$\lim_{max\triangle x_k \to 0} \Sigma_{k=1}^{n} \frac{e^{x_k^*}}{1+x_k^*}$$
as a definite integral with $a = 0$ as the lower limit of integration and $b = 3$ as the unpper limit of integration.

5. Use the properties of the definite integral and appropriate geometric formula to evaluate
   (1) $\int_{-2}^{2} (1-3|x|)dx$ (2) $\int_{-1}^{1} 3 + \sqrt{1-x^2}\ dx$

6. Find $\int_1^5 f(x)\ dx$ if $\int_0^1 f(x)\ dx = -2$ and $\int_0^5 f(x)\ dx = 2$.